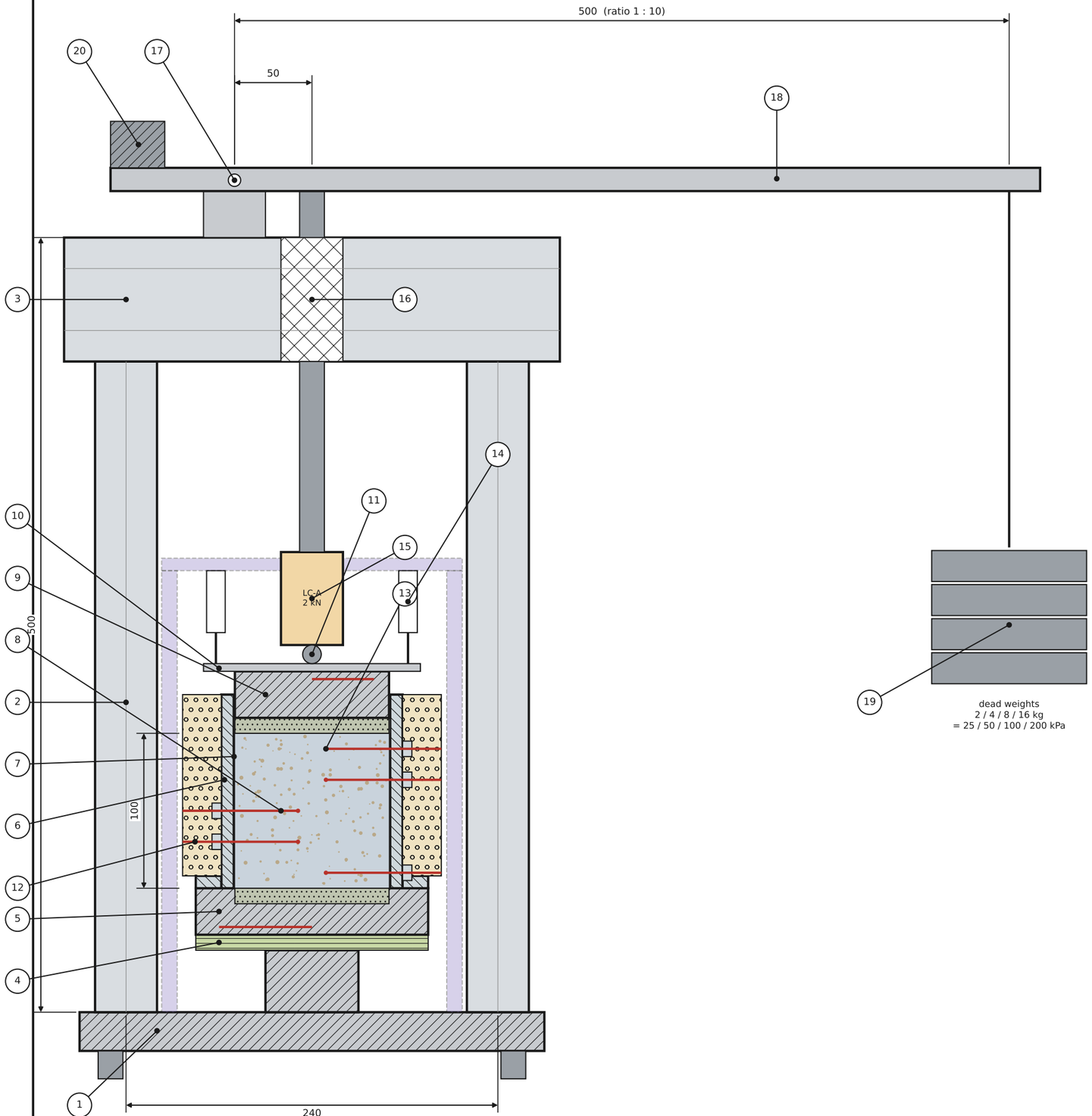


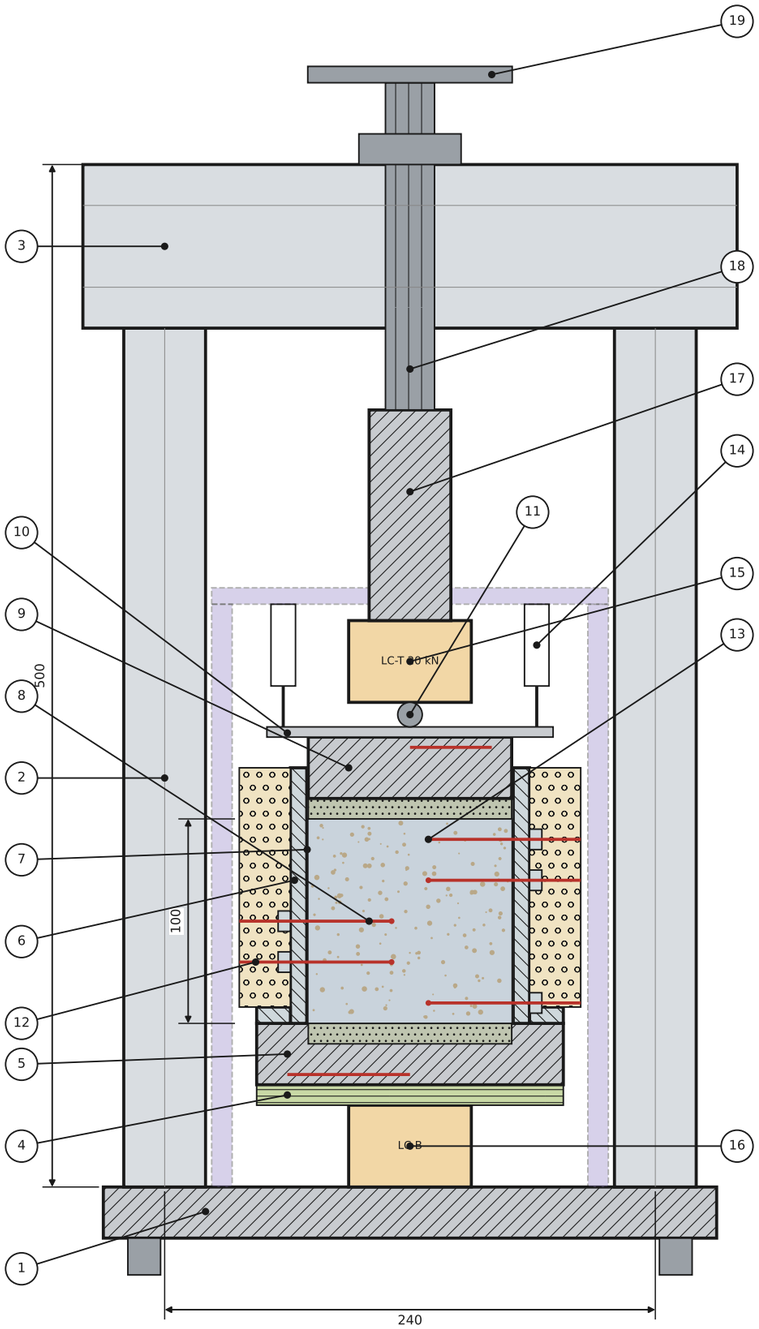
1. base plate Al 6082, 300×250×25, 3 levelling feet
2. columns 2× Al profile 40×80 (strong axis), h 420
3. cross-beam Al profile 80×80, l 320, core-bolted
4. G10/GFRP disc ø150×10 – thermal break, stiff
5. pedestal Al ø150×30 with porous stone ø99.5×10
6. cell PETG-CF ø117/101, foot ø150 (E7-V03), not bolted
7. PTFE liner 0.5 mm + silicone grease (side friction)
8. sample ø100 × 100
9. piston Al ø99.5×40 (gap ≥ 0.3), porous stone below
10. LVDT target plate Al ø140×5 on the piston
11. steel ball ø12 in conical seats (no moment)
12. insulation Kaiflex 25 around the cell
13. thermistors ø1.6 in printed bosses, 5 depths
14. 2× LVDT ±10 mm on Invar reference frame (behind)
15. S-beam load cell 2 kN, C3, comp. −10…+40 °C, rod-ends
16. loading rod ø16 in linear ball bushing (cross-beam)
17. lever pivot: needle bearings / knife edge
18. lever Al 30×15, 600 long, level ± 3°
19. weight hanger + steel discs
20. counterweight (lever self-weight balanced)

Why a lever with dead weight: the load stays constant when the sample heaves or settles and when the temperature changes; no air or power in the freezer. Re-level the lever when heave exceeds ≈ 25 mm at the hanger (2.5 mm at the sample) – turnbuckle on the rod. LC-A logs the real load (pivot friction, lever angle). Alternative: rolling-diaphragm air cylinder with a precision regulator outside the freezer (dry air, closed loop on LC-A).

stress-controlled freeze–thaw test; heave and thaw settlement measured by LVDT

Variant A – constant load (lever, dead weight)

status	DRAFT	drawn	Claude Code (draft)	checked: —
units	mm	project	E7 – freeze–thaw loading tests	date 2026-10-20
scale	**1 : 3**	drawing no.	**E7-V01**	sheet 1 / 5



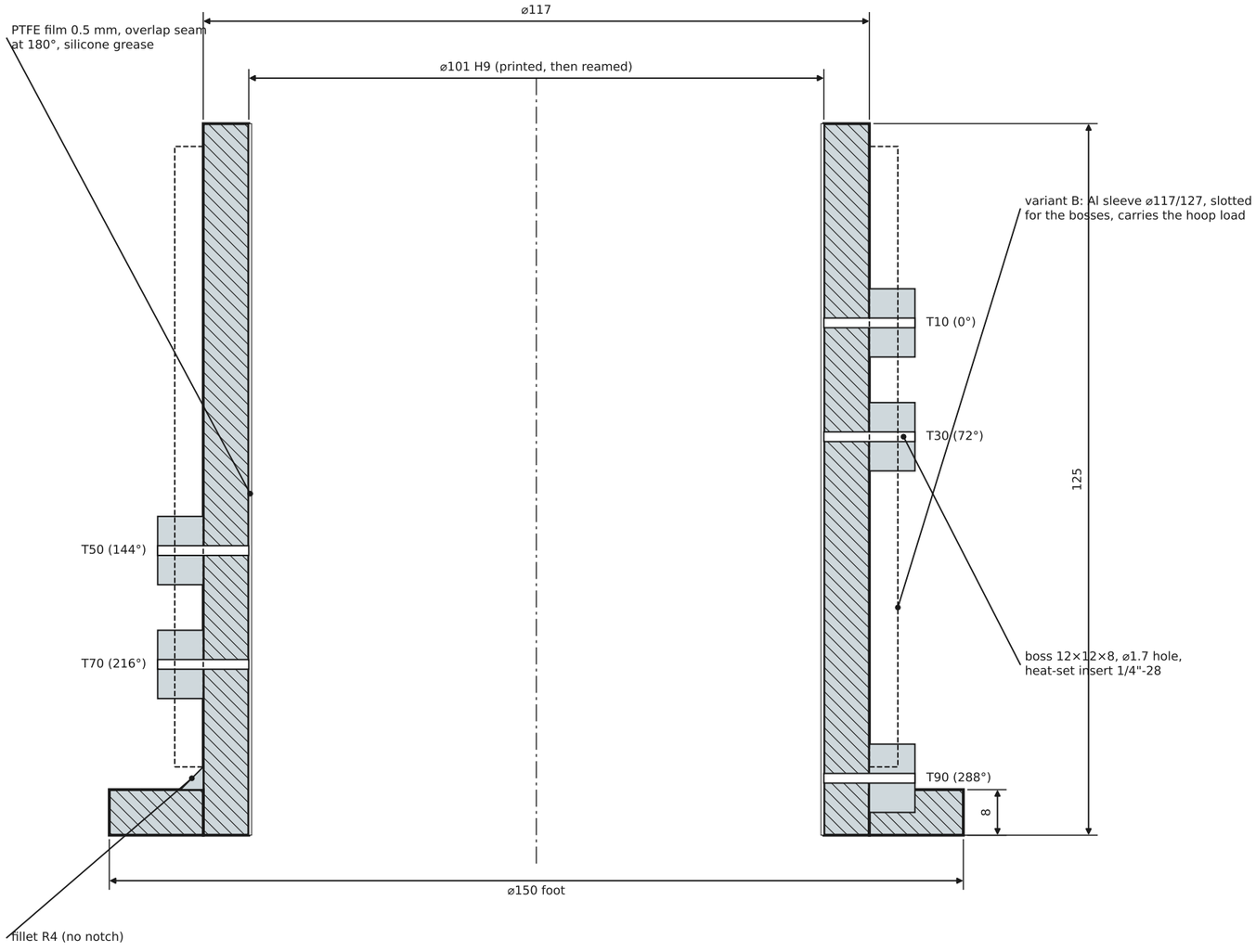
1. base plate Al 6082, 300×250×25, 3 levelling feet
2. columns 2× Al profile 40×80 (strong axis), h 420
3. cross-beam Al profile 80×80, I 320, core-bolted
4. G10/GFRP disc ⌀150×10 – thermal break, stiff
5. pedestal Al ⌀150×30 with porous stone ⌀99.5×10
6. cell PETG-CF ⌀117/101, foot ⌀150 (E7-V03), not bolted
7. PTFE liner 0.5 mm + silicone grease (side friction)
8. sample ⌀100 × 100
9. piston Al ⌀99.5×40 (gap ≥ 0.3), porous stone below
10. LVDT target plate Al ⌀140×5 on the piston
11. steel ball ⌀12 in conical seats (no moment)
12. insulation Kaiflex 25 around the cell
13. thermistors ⌀1.6 in printed bosses, 5 depths
14. 2× LVDT ±10 mm on Invar reference frame (behind)
15. LC-T compression load cell 20 kN, canister, stiff
16. LC-B same type under the stack → wall friction
17. Al spacer tube ⌀40 – thermally matched load path
18. fine-thread screw M24×1.5 + lock nut (set height)
19. handwheel; lock after seating at 5 kPa

Design load 2 MPa ≈ 15.7 kN (silts reach 0.2–2 MPa when restrained).
Constant height is approximate: frame + load cell yield ≈ 0.16–0.29 mm
at 15.7 kN (54–98 kN/mm, vypocty.md §2). LVDTs record the real piston movement; report
pressure together with the measured restraint stiffness.
Temperature artefact: Al columns shrink 0.19 mm for 20 K; with a steel
stack this alone would press ≈ 1.7 kN (≈ 220 kPa) into frozen soil.
→ Al spacer (item 17) matches the columns, and a blank run with a dummy
specimen through the same program gives the correction curve.
Wall friction = LC-T − LC-B (PTFE liner keeps it small).

frost-heave force measured at the top and the bottom; frame stiffness and temperature artefact

Variant B – constant height (restrained heave)

status	DRAFT	drawn	Claude Code (draft)	checked: —
units	mm	project	E7 – freeze–thaw loading tests	date 2026-10-20
scale	**1 : 3**	drawing no.	**E7-V02**	sheet 2 / 5



Material check – PETG-CF

load case	stress	SF short	SF long
hoop stress, A 200 kPa	1.3 MPa	> 30	> 9
hoop stress, B 1 MPa	6.3 MPa	6–9	1.9–2.6
hoop stress, B 2 MPa	12.6 MPa	3–4	1.0–1.3
hoop, B 2 MPa + Al sleeve 5 mm	≈ 1 MPa*	> 40	> 12
axial (Z), friction μ 0.5, 1 MPa	5.8 MPa	2.6–5	< 1.6
axial (Z), PTFE μ 0.05, 1 MPa	0.6 MPa	> 25	> 7

Assumed strength (data-sheet range, verify): XY 40–55 MPa, Z 15–30 MPa, E 3–6 GPa; long-term limit ≈ 30 % (creep, layer bond). Details: vypocty.md §4.

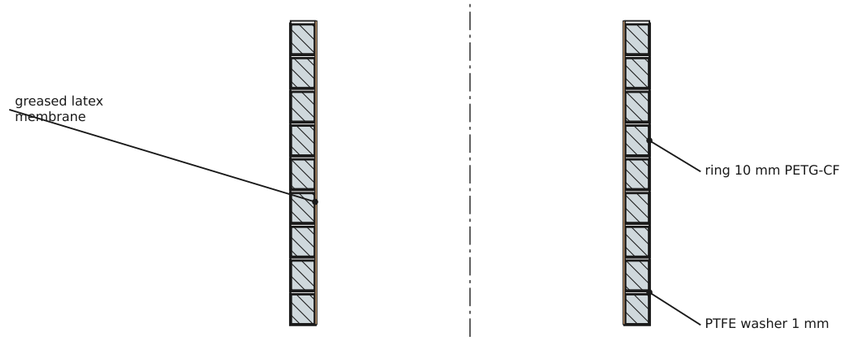
Verdict: suitable for variant A; for B (design 2 MPa) only with the Al sleeve shown dashed (*it carries ≈ 90 % of the hoop load: 70 GPa×5 mm vs. ≈ 4 GPa×8 mm), the PETG-CF then acts as a liner.
Conditions for both:
- the wall is ≥ 8 mm and printed with concentric perimeters only (100 % infill, no zig-zag), so the hoop stress runs along the filament;
- friction is kept low (PTFE liner) and the cell is not clamped at the foot, so friction pushes the wall instead of pulling the layers apart;
- the inside is sealed (thin epoxy coat) so water cannot enter print voids and split layers when it freezes;
- there are no printed threads or sharp notches (CF lowers toughness at −10 °C) – heat-set inserts and fillets instead.

Variant A: a multi-ring cell (view F) is the literature standard against wall friction – printed rings 10 mm + PTFE washers + greased latex.

Print settings (starting point)
hardened nozzle 0.6 mm, layer 0.2 mm, 250–260 °C, bed 80 °C, slow outer walls, filament dried 4 h at 65 °C, enclosure, part cooling ≤ 30 %.

Qualification before use
1 ring burst test: printed ring 20 mm high, internal water pressure to failure (target ≥ 3× design pressure);
2 10 freeze–thaw cycles −10/+10 °C of water-saturated coupons, then tensile test XY and Z vs. dry control;
3 creep: ring under 1 MPa for 7 days at +10 °C, hoop strain gauge.

VIEW F – multi-ring cell (variant A) 1 : 2

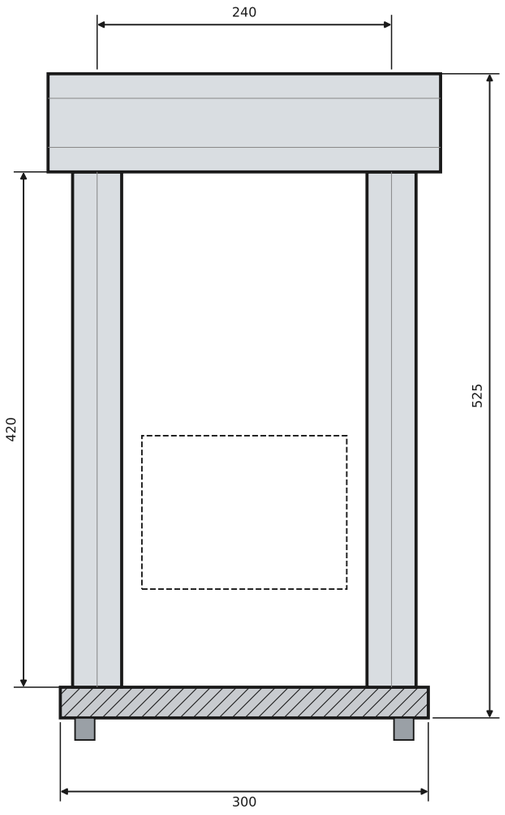


print upright; hoop direction = layer direction; foot rests on the pedestal, not bolted

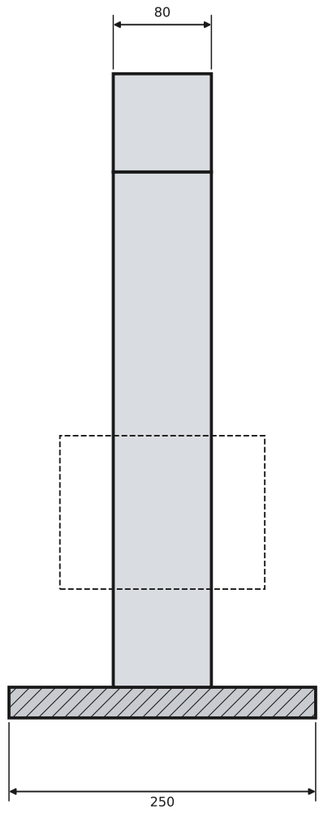
Sample cell – PETG-CF, 3D printed

status	DRAFT	drawn	Claude Code (draft)	checked: —
units	mm	project	E7 – freeze–thaw loading tests	date 2026-10-20
scale	**1 : 1**	drawing no.	**E7-V03**	sheet 3 / 5

FRONT VIEW 1 : 5



SIDE VIEW 1 : 5



Bill of materials

#	part	size	qty
1	base plate Al 6082 T6	300 × 250 × 25	1
2	profile 40×80 (column)	l = 420	2
3	profile 80×80 (cross-beam)	l = 320	1
4	core bolt M12 into tapped profile end		8
5	socket screw M10 base → column end		4
6	levelling foot M12, steel		3
7	Invar rod ⌀10 + Invar/Al bridge (LVDT)	l = 300	2 + 1
8	linear ball bushing ⌀16 (A) / nut M24×1.5 (B)	in cross-beam	1
9	end caps, T-slot covers		–

Profile values used in vypocty.md §2 are typical catalogue values (series 8): 40×80 A ≈ 1050 mm², 80×80 I ≈ 1.4·10⁶ mm⁴ – check against the chosen supplier.

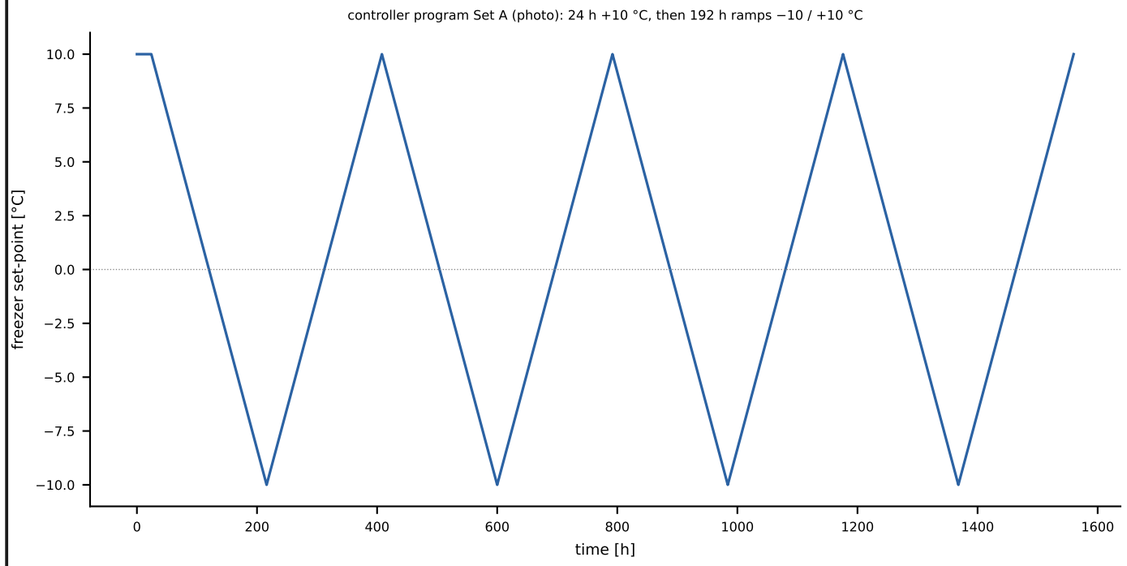
Stiffness at 15.7 kN (2 MPa): cross-beam 0.046 mm, columns 0.045 mm, load cell ≈ 0.05 mm, joints 0.02–0.15 mm. Joints dominate → core bolts into tapped profile ends, torque-controlled, no angle brackets; optionally 2 steel tie rods M12 next to the cell.

Fit in the chest freezer: frame 300 × 250, height 525 mm incl. base; variant A needs the lever (≈ 600 mm long, 120 mm above the beam) – check against the inside height (≈ 650 mm, to be measured).

one frame for both variants; core-bolted joints, no angle brackets in the load path

Frame – aluminium profiles

status	DRAFT	drawn	Claude Code (draft)	checked: —
units	mm	project	E7 – freeze–thaw loading tests	date 2026-10-20
scale	**1 : 5**	drawing no.	**E7-V04**	sheet 4 / 5



Weighing load cells or force transducers?
Use force transducers (axial tension/compression load cells), not platform/single-point weighing cells:
- they take a centred axial force in series in the load path; weighing cells are made for a platform with off-centre loads and bend;
- compression canister / column cells are much stiffer (≈ 0.02–0.1 mm at full scale) – essential for constant height (variant B);
- S-beams give fine resolution at low loads (variant A, 2 kN) and take rod-ends that keep moments out;
- choose a temperature-compensated range covering −10 °C (typ. −10…+40 °C), stainless, IP67; log the cell temperature;
- load introduction through a steel ball in conical seats.

A hand-held force gauge is not suitable for long logged tests.

Readout: full bridge, 6-wire, ratiometric on the CR3000 (or a 24-bit bridge amplifier); 2.5 V excitation; zero before each run at +10 °C.

Blank run (both variants)
Run the full program with an Al dummy specimen ⌀100×100 instead of soil. The recorded force and displacement are the apparatus response to temperature alone (frame contraction, load-cell drift, LVDT drift) and are subtracted from the soil runs.

Derived quantities
heave s = mean(LVDT-1, -2) − blank; heave ratio s / H;
A: s(t) at constant σ; thaw settlement; B: σ_h = LC-T / A;
friction = LC-T − LC-B; frost-front depth from T10…T90 (0 °C crossing).

Channels

ID	sensor	variant	quantity
LC-A	S-beam 2 kN, full bridge 6-wire	A	applied load
LC-T	compression 20 kN	B	heave force at the top
LC-B	compression 20 kN	B	force at the base → friction = T − B
LVDT-1, -2	±10 mm, cold-rated	A, B	heave / settlement, tilt
T10…T90	NTC ⌀1.6 in bosses, r 9	A, B	freezing front, profile
Ttop, Tbot	NTC in piston / pedestal	A, B	boundary temperatures
TLC-T, TLC-B	NTC taped on load cells	A, B	zero-drift correction
Tframe	NTC on column	A, B	frame contraction
Tair	NTC in shield	A, B	freezer air
SG-1, -2 (opt.)	hoop strain gauges on cell	B	lateral pressure (calibrated)

channels for both variants; program = existing controller 'Set A'

Instrumentation and freeze–thaw program

status	DRAFT	drawn	Claude Code (draft)	checked: —
units	mm	project	E7 – freeze–thaw loading tests	date 2026-10-20
scale	–	drawing no.	**E7-V05**	sheet 5 / 5